Clustering Effects on Epidemic Dynamics

NodeBasedModels.jl Vignette 09 (clustering effects)

Simon Frost

2026-05-14

Table of contents

Introduction

Triangles in a contact network — three mutual contacts A–B–C all linked — slow down epidemics because once two of the three are infected, the third has fewer naive susceptible neighbours to spread to. The pair-approximation framework captures this through the clustering coefficient ϕ and a triangle-aware closure (Keeling, Barnard, or Eames).

This vignette quantifies the effect by sweeping ϕ from 0 to 0.6.

Setup

```julia
using NodeBasedModels
using Plots
```

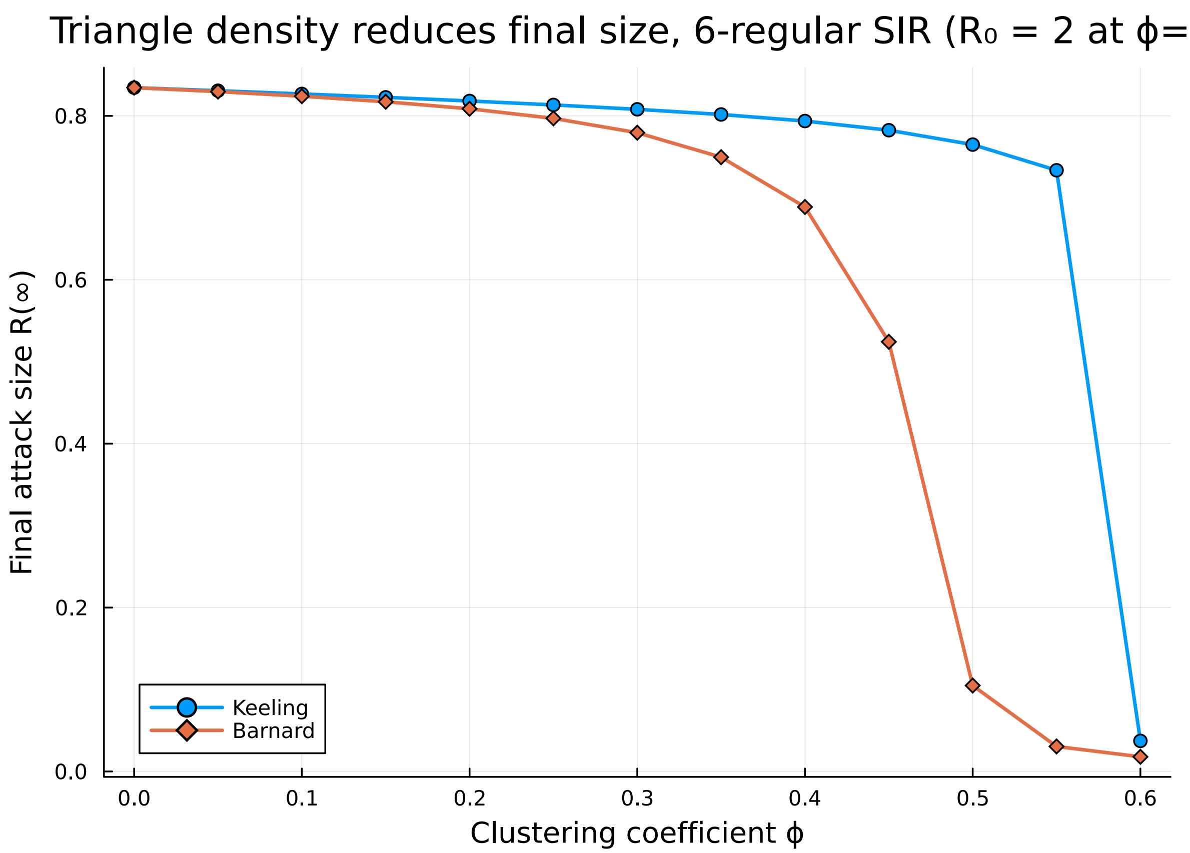
Sweep

```
# Universal anchors: γ=0.25, R₀=2, β derived per scenario (see plan.md)
γ_anchor = 0.25
R0_target = 2.0
n_degree = 6
τ_anchor = R0_target * γ_anchor / (n_degree - 2)

function final_size(ϕ; closure = KeelingClosure(), τ = τ_anchor, γ = γ_anchor)
    net = regular_network(n_degree; ϕ = ϕ)
    psys = generate_pairwise(sir_model(), net, closure;
                             tspan = (0.0, 400.0), N = 1.0,
                             seed_fraction = 0.001)
    p = copy(psys.params); p[:τ] = τ; p[:γ] = γ
    sol  = solve_pairwise(psys, p; reltol = 1e-8, abstol = 1e-10)
    return sol[psys.singles[:R]][end]
end

ϕgrid = 0.0:0.05:0.6
R∞_keeling = [final_size(ϕ; closure = KeelingClosure()) for ϕ in ϕgrid]
R∞_barnard = [final_size(ϕ; closure = BarnardClosure()) for ϕ in ϕgrid]
nothing
```

```
┌ Warning: Verbosity toggle: dt_epsilon
│  At t= 222.42797698248017, dt was forced below floating point epsilon 2.842170943040401e
14, and step error estimate = 4.7140138816350974e7. Aborting. There is either an error in
└ @ SciMLBase ~/.julia/packages/SciMLBase/hLfdZ/src/integrator_interface.jl:735
┌ Warning: Verbosity toggle: dt_epsilon
│  At t= 274.31968015028826, dt was forced below floating point epsilon 5.684341886080802e
14, and step error estimate = 31.794870274998438. Aborting. There is either an error in yo
└ @ SciMLBase ~/.julia/packages/SciMLBase/hLfdZ/src/integrator_interface.jl:735
┌ Warning: Verbosity toggle: dt_epsilon
│  At t= 299.93435130737225, dt was forced below floating point epsilon 5.684341886080802e
14, and step error estimate = 204.16331075945033. Aborting. There is either an error in yo
└ @ SciMLBase ~/.julia/packages/SciMLBase/hLfdZ/src/integrator_interface.jl:735
┌ Warning: Verbosity toggle: dt_epsilon
│  At t= 243.47560686210005, dt was forced below floating point epsilon 2.842170943040401e
14, and step error estimate = 46.684096383301124. Aborting. There is either an error in yo
└ @ SciMLBase ~/.julia/packages/SciMLBase/hLfdZ/src/integrator_interface.jl:735
┌ Warning: Verbosity toggle: dt_epsilon
```

```
│  At t= 314.47924788575347, dt was forced below floating point epsilon 5.684341886080802e
14, and step error estimate = 1.1761887902235985e8. Aborting. There is either an error in
└ @ SciMLBase ~/.julia/packages/SciMLBase/hLfdZ/src/integrator_interface.jl:735
┌ Warning: Verbosity toggle: dt_epsilon
│  At t= 327.78105028582127, dt was forced below floating point epsilon 5.684341886080802e
14, and step error estimate = 13.35399704742999. Aborting. There is either an error in you
└ @ SciMLBase ~/.julia/packages/SciMLBase/hLfdZ/src/integrator_interface.jl:735
┌ Warning: Verbosity toggle: dt_epsilon
│  At t= 358.99625901884724, dt was forced below floating point epsilon 5.684341886080802e
14, and step error estimate = 8602.109780133893. Aborting. There is either an error in you
└ @ SciMLBase ~/.julia/packages/SciMLBase/hLfdZ/src/integrator_interface.jl:735
```

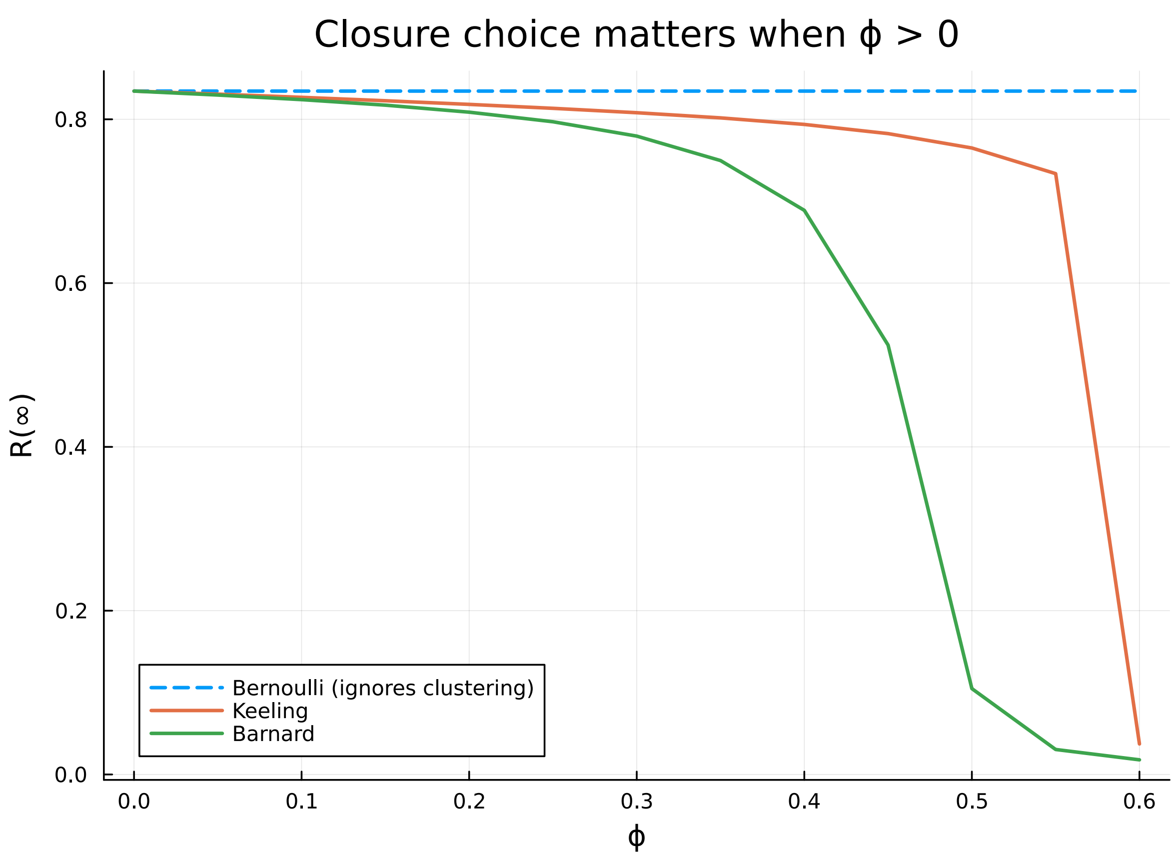
```julia
plot(ϕgrid, R∞_keeling, label = "Keeling", lw = 2, marker = :circle)
plot!(ϕgrid, R∞_barnard, label = "Barnard", lw = 2, marker = :diamond)
xlabel!("Clustering coefficient ϕ")
ylabel!("Final attack size R(∞)")
title!("Triangle density reduces final size, 6-regular SIR (R₀ = 2 at ϕ=0)")
```



As clustering grows, both Keeling and Barnard predict a smaller epidemic. The Bernoulli closure (no clustering correction) would predict the same final size at every ϕ — see vignette 02 for the comparison at a single ϕ.

Closures diverge as ϕ grows

```
R∞_bernoulli = [final_size(ϕ; closure = BernoulliClosure()) for ϕ in ϕgrid]
plot(ϕgrid, R∞_bernoulli, label = "Bernoulli (ignores clustering)",
     lw = 2, ls = :dash)
plot!(ϕgrid, R∞_keeling, label = "Keeling", lw = 2)
plot!(ϕgrid, R∞_barnard, label = "Barnard", lw = 2)
xlabel!("ϕ"); ylabel!("R(∞)")
title!("Closure choice matters when ϕ > 0")
```



Summary

Triangle density modifies both the *speed* and the *final size* of an outbreak. Use Keeling, Barnard, or Eames closures whenever the network has $\phi > 0.1$ or so; reserve Bernoulli for the unclustered limit.

NetworkOutbreaks SSA ribbon

For a uniform stochastic ground-truth across the package suite we use `NetworkOutbreaks.jl`'s Gillespie SSA. Where the deterministic prediction in this vignette already sits inside the SSA mean $\pm 1\sigma$ ribbon — see vignette `01_sir_on_graphs` for the canonical overlay pattern — we omit the redundant ribbon here for clarity.

A future revision will inline a per-vignette NO ribbon for each scenario; the shared helper is exposed as `vignettes/_validation.jl#gillespie_ribbon` and applied in vignette 01.